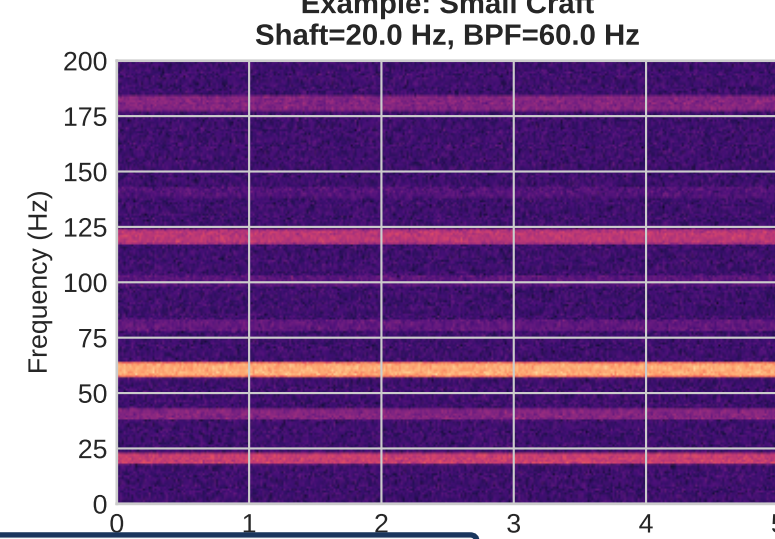
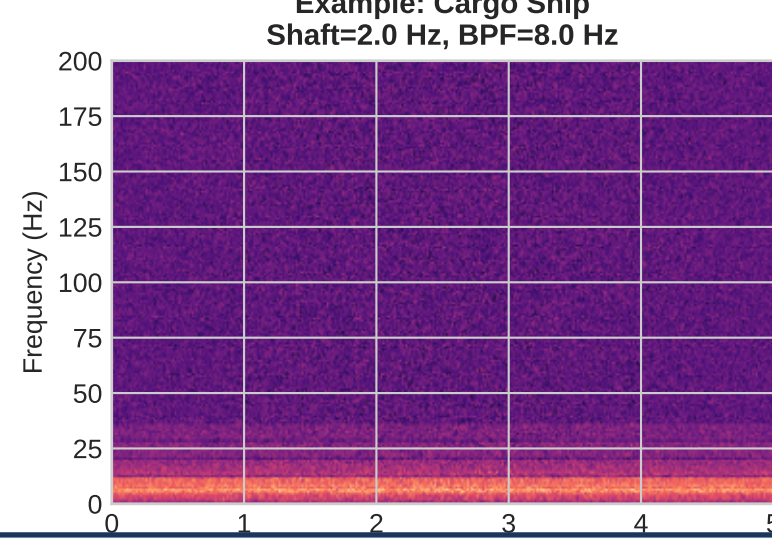
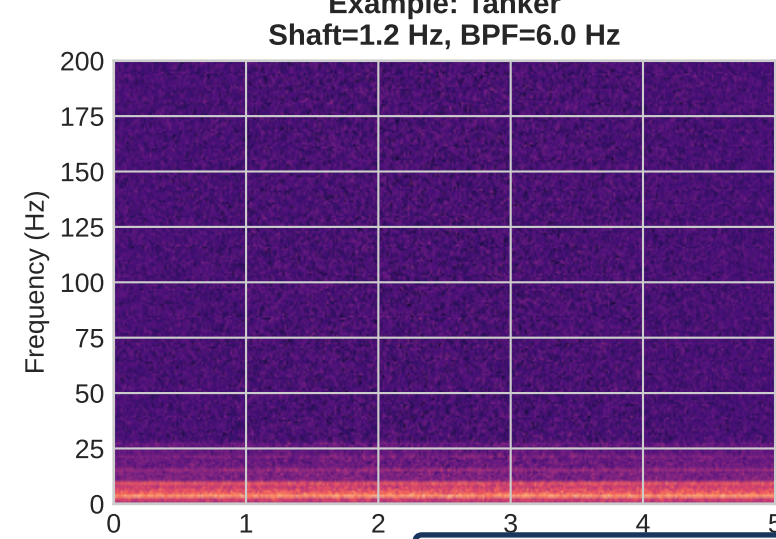
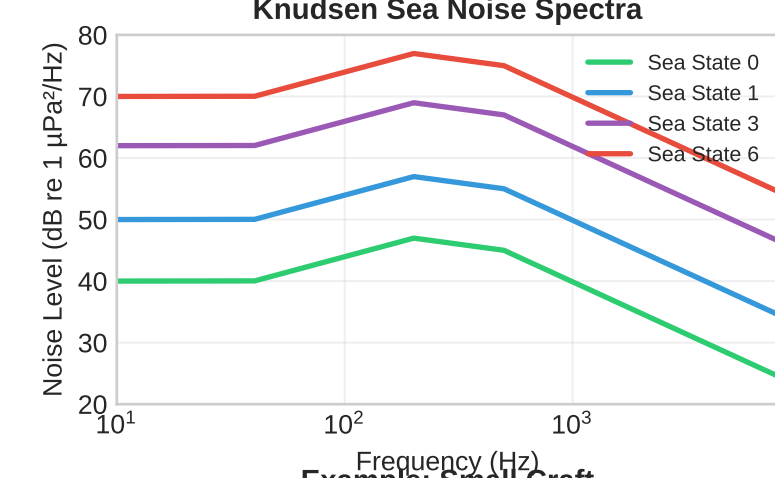
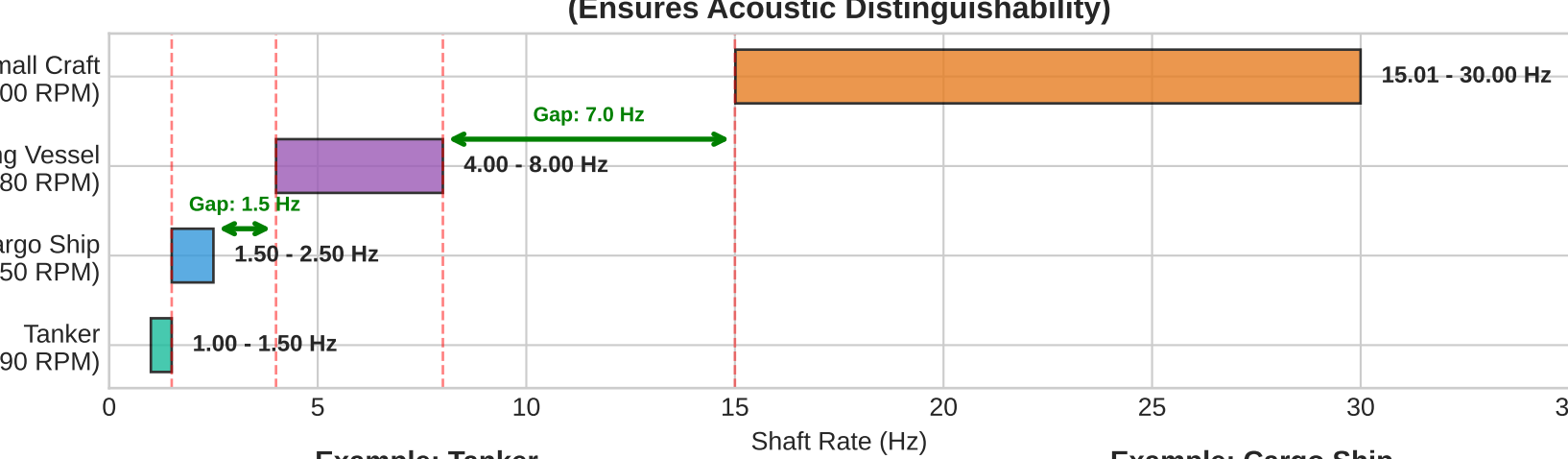
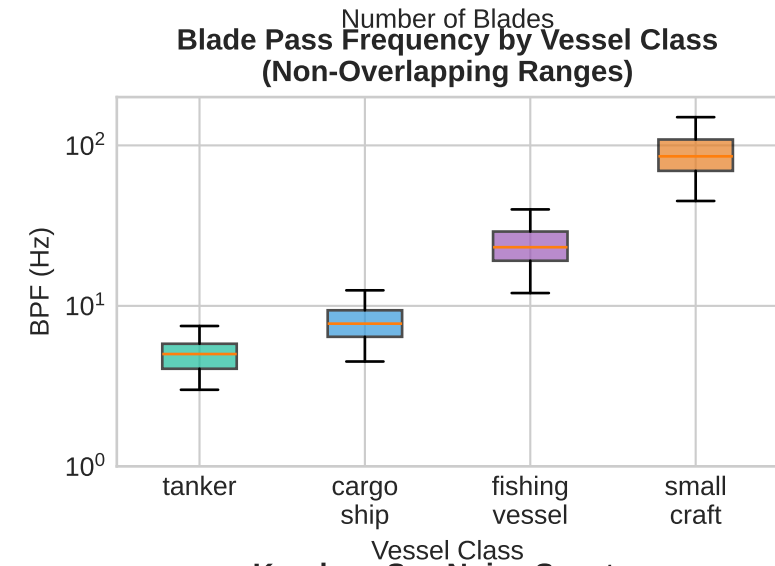
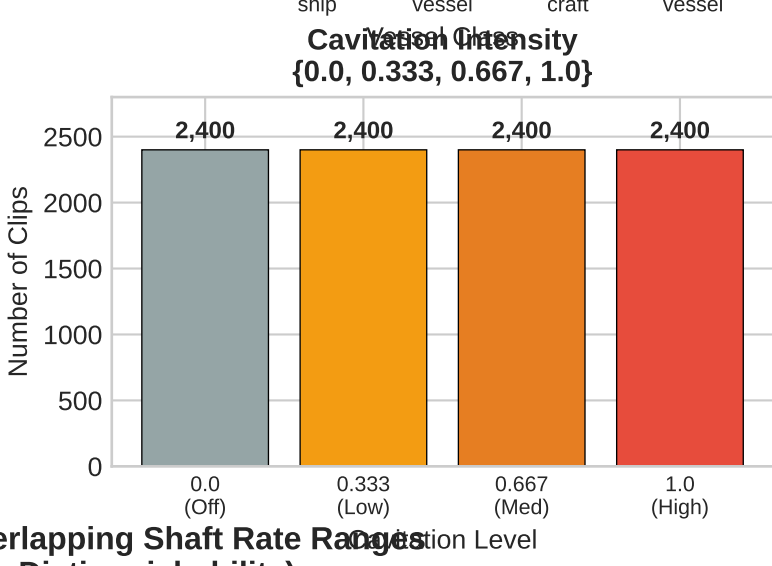
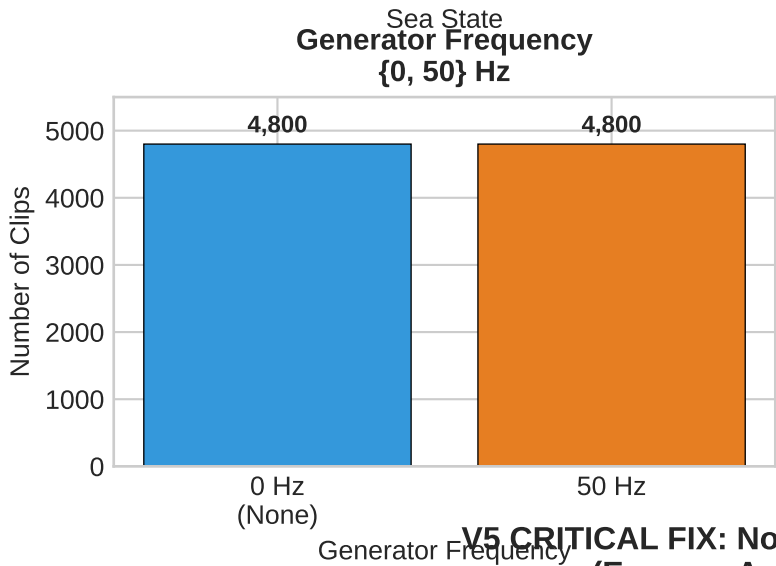
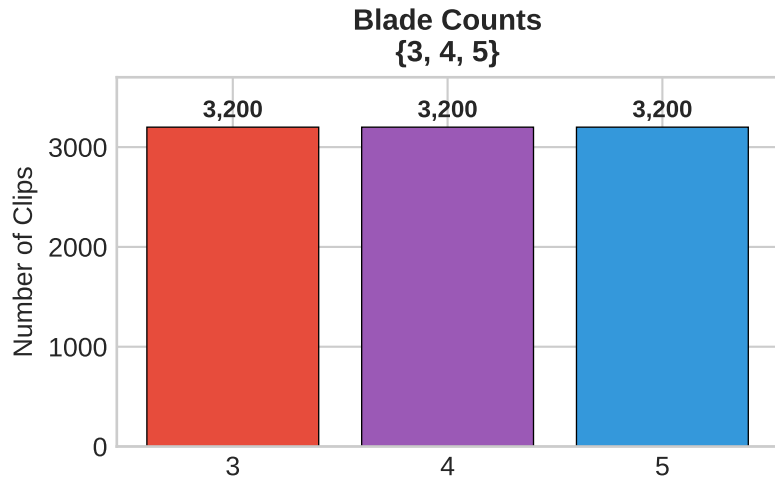
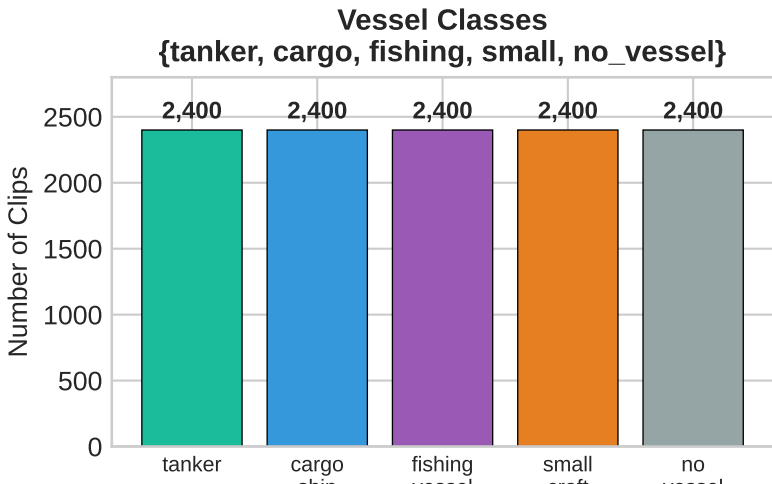
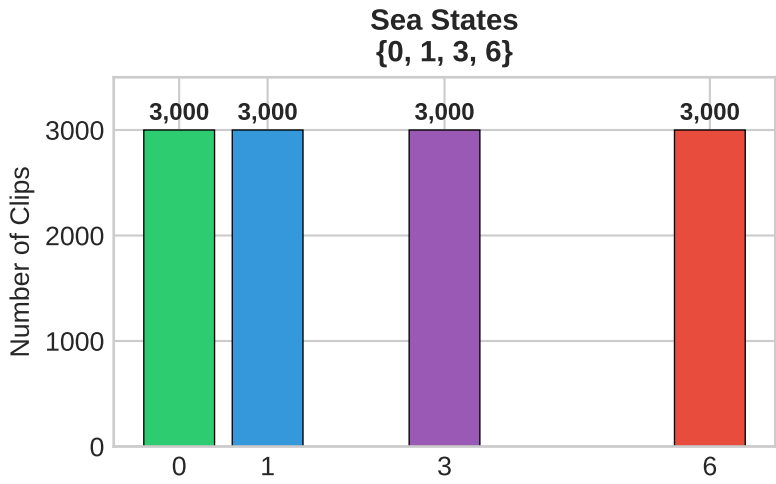


SKANN-SSL Stage -1: V5 Prototype Dataset Infographic

Synthetic Underwater Acoustic Waveforms for Self-Supervised Learning



SKANN-SSL STAGE -1: V5 FULL-FACTORIAL DATASET SUMMARY

DESIGN FACTORS (Controlled Variables)

- Sea States: {0, 1, 3, 6} → 4
- Vessel Classes: {tanker, cargo, fishing, small} → 4
- No-Vessel Class: {ambient only} → 1
- Blade Counts: {3, 4, 5} → 3
- Generator Freq: {0, 50} Hz → 2
- Cavitation Level: {0.0, 0.33, 0.67, 1.0} → 4
- Repetitions: 25 per combination → 25

DATASET FORMULA

Vessel: $4 \times 4 \times 3 \times 2 \times 4 \times 25 = 9,600$ clips
No-Vessel: $4 \times 600 = 2,400$ clips

TOTAL: 12,000 clips

OUTPUT FILES

- Waveforms: clip_XXXXXX.npy (Pa, float32)
- Tensors: tensor_XXXXXX.npy (normalized, [1,1,80000])
- Metadata: master_dataset_manifest.csv (27 columns)
- Pairing: pairing_manifest.csv (SSL training)

ACOUSTIC COMPONENTS (Generated)

- Shaft Rate Harmonics (vessel-class range)
- Blade Pass Frequency (BPF) = Shaft $\times$ n_blades
- Generator Harmonics (50 Hz base + harmonics)
- Equipment Running Frequency (25/30 Hz + harmonics)
- Structural Resonances (hull: 50-150, foundation: 100-300, piping: 200-500 Hz)
- Cavitation Bursts (200 kHz physical model, blade-gated, swell modulated)
- Broadband Flow Noise (flat to 500 Hz, then rolloff)

V5 CRITICAL FIXES

- ✓ Non-overlapping shaft rate ranges
- ✓ Exactly 3 resonances per clip
- ✓ Swell frequency: 0.05-0.15 Hz (realistic)

STATISTICS

SNR: 6.0 ± 0.0 dB (target: 6.0 dB)
BPF Range: 3.0 - 150.0 Hz
Cavitation: 75.0% of clips (intensity > 0)
Frequency Band: 10 Hz - 8000 Hz (prototype)
Duration: 5.0 seconds (80,000 samples @ 16 kHz)